

verity

Technical solution document

How the Stylified workspace is built, hosted, secured, backed up and tested.

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DOCUMENT

5 of 6

How it is built

For Stylified's owner and for anyone you ask to review the solution. It says what exists, how it is checked, and what is not built.

01 Overview

Stylified gets its own workspace on Verity's hosted platform. The workspace is a set of screens and rules for the piece-rate cycle, on top of a shared foundation that provides sign-in, roles, an audit trail, files, notifications and company isolation.

EXPERIENCE	Web application for desktop, tablet and phone, dark and light themes. Supervisor screens are designed phone-first and open from a camera scan. Hindi and English labels on the floor screens, daily entry, slips, and the munshi's and store's daily screens.
PIECE-RATE MODULE	Fabric rolls, lots and rate cards, bundles, stages, karigars and rates, handovers and returns, rework, cases, wages and money, lot costing, approvals, day close, team, reports. Every change is a *command* with its own checks; every screen reads through a *query*.
PLATFORM FOUNDATION	Company isolation, identity and memberships, roles built from permissions, an audit trail, an event log, file storage, notifications, numbering, configuration.
DATA AND HOSTING	A relational database with row-level security forced on every Stylified table, file storage, application hosting in an Indian region, backups.

02 Hosting

PART	WHERE AND HOW
Application	Hosted serverless application, deployed to the Mumbai region (configured in the deployment settings)
Database	Managed PostgreSQL in the Mumbai region, reached through a connection pooler in transaction mode. The application connects as a restricted role that cannot bypass row-level security
Files	Object storage with the same company isolation; used for photographs
Sign-in	The platform's identity service; passwords never reach Verity's own tables

PART	WHERE AND HOW
Environments	A demonstration workspace with fictional data and Stylified's own workspace are separate tenants. Nothing is shared between them
Deployment shape	Verity's shared hosted deployment (one application and database for many companies, each isolated as in section 3). A dedicated instance on Stylified's own Vercel and Supabase accounts is **not part of this solution**: see below

If Stylified later wants its own accounts

Running Verity on a customer's own Vercel and Supabase accounts is feasible but not yet built or validated. Our research lists what must be done first: scripting the creation of the restricted database role the application connects as, a scripted first-operator and first-company setup on a fresh Supabase project, a scheduled-jobs setup per instance, separate file and database backups, a release and migration channel for more than one instance, and a support model without a shared console. A rehearsal on throwaway accounts must pass before any real data moves. Provider plans must allow business use (Vercel Pro and Supabase Pro are the minimum; free plans are not suitable). We will quote the move separately once this is done; until then the isolation, backup and security statements in this document describe the shared hosted deployment only.

Provider names and plans can change over the life of the service; the data stays in an Indian region and the isolation rules below do not change.

03 Company isolation

- Every Stylified record carries the company it belongs to. The database itself refuses to return or accept a row for another company: row-level security is **forced** on each table and the company is set for each request from the signed-in user's verified membership, **never from anything the browser sends**.
- If no company is set, a query returns nothing and a write is rejected: isolation fails closed.
- The platform checks at startup that its database role is one that cannot bypass these rules, and refuses to run otherwise. An automated test proves that one company cannot read another's data.

04 Roles and permissions

- A permission is an action on a kind of record within a scope (for example "issue" on a piece-rate entry for the company). A role is a named set of permissions. There are no hard-coded role names in the code, so changing a role is a setting, not a release.
- The six Stylified roles (Owner, Manager, Supervisor, Munshi, Store, Cutting master) and their matrix are in document 2, section 4. They are installed once per workspace from a preset and can be edited.
- The server checks the permission for every command and every query. Hiding a button is only a convenience; it is never the control.
- **Supervisor scope.** The platform's scope list stays as it is. A supervisor's assigned stages are held as a small record in the piece-rate module, and a karigar can be set under a supervisor. The give-out and take-back commands refuse a supervisor acting on a stage that is not theirs, and a give-out to a karigar who is set under another supervisor. This is tested with a second supervisor.

05 The data model

RECORD	WHAT IT HOLDS	RULES
Stage	A step in the line, in order, with a default rate per piece, an active flag and whether it is required or optional (a lot can skip only an optional stage)	Stages are Stylified's data; they can be renamed and reordered
Roll and roll movement	Roll number, fabric, unit (metres or kilograms), quantity, supplier, invoice, rate, and every movement in or out	Append-only movements; the balance is worked out from them; a repeated invoice is warned
Lot and lot rate card	Number, name, design, planned pieces, dates, status; a rate per stage for this lot, and a stage can be skipped	Numbers are gapless and never reused; a closed lot is locked. The lot's rate is used before the worker's, then the stage default
Bundle	Number B-0001 onward, net pieces, size-mix note, the lot and the rolls it was cut from	Numbers are gapless per workspace; bundles can be split or joined at the cutting table until work has left it
Karigar or unit	Name, kind, contact, address, joined on, ID note, emergency contact, the supervisor, active	Retired, never deleted
Rate	Rupees per piece for a karigar at a stage, the wage basis, the date it applies from	History is kept; each entry takes a snapshot of the rate used
Entry (issue or return)	Bundle, stage, karigar, pieces sent and counted, due date, confirmation method, challan number, and for a return the split into good, rejected, rework and short	Append-only. A database check makes the four parts sum to the pieces returned. Corrections are reversals
Payment	Advance, recovery, deduction, payment or opening balance, amount, mode, reference, reason	Append-only; amounts are whole paise, so there is no rounding drift
Lot cost	A cost in one of your categories against a lot, with the date and who entered it	Append-only; a cost named like fabric, wages or a stage is refused (already counted)
Ready stock move	Opening stock (named, no lot), stock out with the party and reference, and count corrections with a reason; what production made ready is derived from the bundles	Append-only; a reversal is the only undo; nothing can leave that is not in stock
Stage assignment	A supervisor and a stage	Set by the owner; checked by the commands
Approval and case	Kind (deduction, write-off, hold, close, charge), subject, amount or pieces, requester, state, decision and reason; a case ties a shortage to the worker who held the pieces and the supervisor who recorded it	Decided once; above-limit requests need the owner; nobody decides their own request
Evidence photo	A photograph about a rejected or rework entry; lot pictures set up with you	Stored with company isolation; never a public link

RECORD	WHAT IT HOLDS	RULES
Day close	Supervisor, date, summary figures at the time, a note	Frozen once submitted
Settings and reasons	Manager limits, the lost-piece charge, the low-fabric level, the challan heading, reasons and cost categories in two languages	Your data, editable on screen

How the figures are made

Nothing on a dashboard is typed in. Pieces out, pieces waiting, good, rejected, rework, short, days at a stage, roll balances, wages, balances and lot costs are all computed from the entries, movements and payments by one piece of logic, which is covered by automated tests that use the worked examples from your requirement sheet (for example 100 pieces issued, 94 good, 2 rejected, 1 rework, 2 short and 1 still open; and ₹48,000 over 960 good pieces is ₹50 a piece). A figure on a report always agrees with the entries it is built from.

06 Audit and corrections

- Every command records who did it, when, from which role, and what changed, in an audit trail the owner can read.
- Entries, payments and rate history cannot be updated or deleted by the application: a database rule rejects both. A wrong entry is cancelled by a **reversal** with a reason; the original and the reversal both stay visible.
- A reversal that would leave a later stage with more pieces than it received is refused.

07 Photographs

- A photograph is uploaded from the phone camera or a file, checked for type and size, stored in the company's isolated storage, and shown through a short-lived link only to users who may see the record.
- Pictures are resized for the screen so lists stay fast.

08 Backups and recovery

- A daily backup of the database and files, kept for at least 30 days, is part of the service (document 3, section 5). The mechanism (the hosting provider's backups plus our own nightly export) is configured and a **restore is rehearsed** on Stylified's workspace before go-live; that rehearsal is an acceptance criterion.
- Because corrections are reversals, a restore is needed only for loss or damage, not for ordinary mistakes.

09 Security summary

CONCERN	CONTROL
One company seeing another's data	Forced row-level security in the database; company taken from the verified membership; tested

CONCERN	CONTROL
A user doing more than their role allows	Server-side permission check on every command and query; tested
Altering history	Append-only ledgers enforced by the database; reversal only
Staff access to your data	Only to support you; each session recorded in your audit trail and time-limited
Secrets	Held in the hosting environment, never in code or in the database
Public links	The bundle QR opens a sign-in screen, then the bundle. It carries only the bundle's address, no figures or names. Roll and challan labels work the same way

10 Speed

Measured on a production build of the demonstration workspace with the database beside the application: a page is made on the server in 50 to 80 milliseconds and is idle in the browser in 0.7 to 1.0 seconds. On a year of generated history (24,000 entries over 120 lots) the factory dashboard takes about 0.5 seconds, the lot list 0.5 seconds and the supervisor's screens about 50 milliseconds. On the hosted service the network and the sign-in check add to these; we will report measured times on Stylified's workspace before go-live, not estimates.

11 Testing

LAYER	WHAT IS COVERED
Logic tests	The piece-rate calculations, the spreadsheet writer and the sign-in address rules, using the requirement sheet's own examples
Database tests	More than 85 tests in all against a real database: stage and karigar scope for supervisors, approvals within the owner's limits, the frozen day close, reconciliation, over-issue and over-return refusals, rework pool, the wage bases and the lot rate card, the receiver's count with cases and charge requests, rolls and fabric used by a cut, lot costing and its refusals, ready stock in and out, required and optional stages, where each role opens, role presets that cannot grant more than the installer holds, opening balances, append-only rules, company isolation
Platform tests	Identity, membership, tenant isolation and permission tests that every capability inherits
Browser checks	Every route opened as each role in dark and light on desktop and at 390 px, with the demonstration data, before go-live; defects are logged and fixed
Acceptance	The criteria in document 2, section 10, run with Stylified

12 Where things stand, and what follows

- The piece-rate module is built and proven on demonstration data in our test environment; it moves onto Stylified's own workspace and people at kickoff.
- Speed on the hosted service is measured on your workspace during the week-three checks; the figures above are from a local build.
- The hosted run (photograph storage, a supervisor on a real phone) and the restore rehearsal happen on your workspace before go-live and are acceptance criteria.
- The Zebra label file (ZPL) is written to the printer's specification and checked as text; it has not been run on a printer, because the model is not chosen.
- Not built: Hindi on lots, costs, reports and settings, a TSC label file, packing and dispatch, WhatsApp or SMS, offline use, a Tally export. They are quoted separately.
- A deployment on Stylified's own Vercel and Supabase accounts is a later option, offered after a full rehearsal.